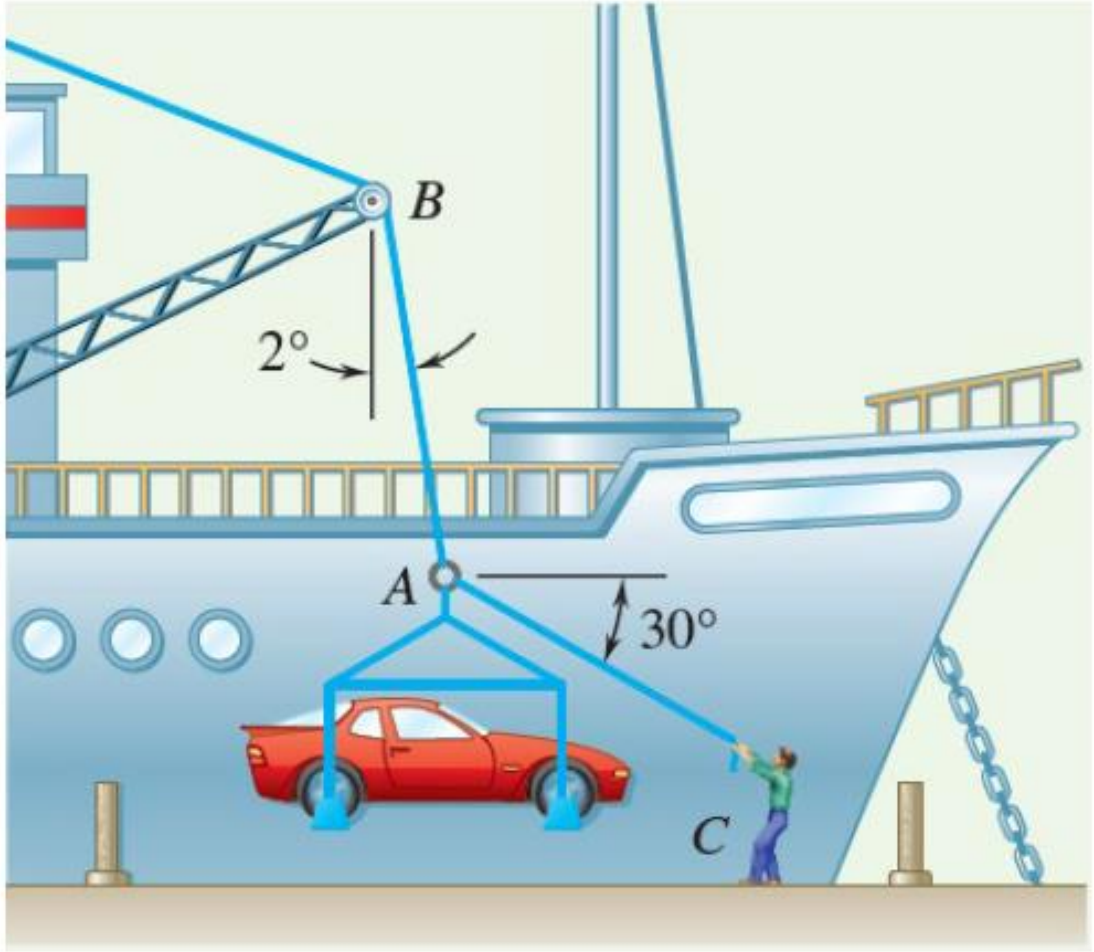
In a ship-unloading operation, a 3500-kg automobile is supported by a cable. A worker ties a rope to the cable at A and pulls on it in order to center the automobile over its intended position on the dock. At the moment illustrated, the automobile is stationary, the angle between the cable and the vertical is 2°, and the angle between the rope and the horizontal is 30°. What are the tensions in the rope and cable?



$T_{AB} = 3570$ ~~lb~~ Kg

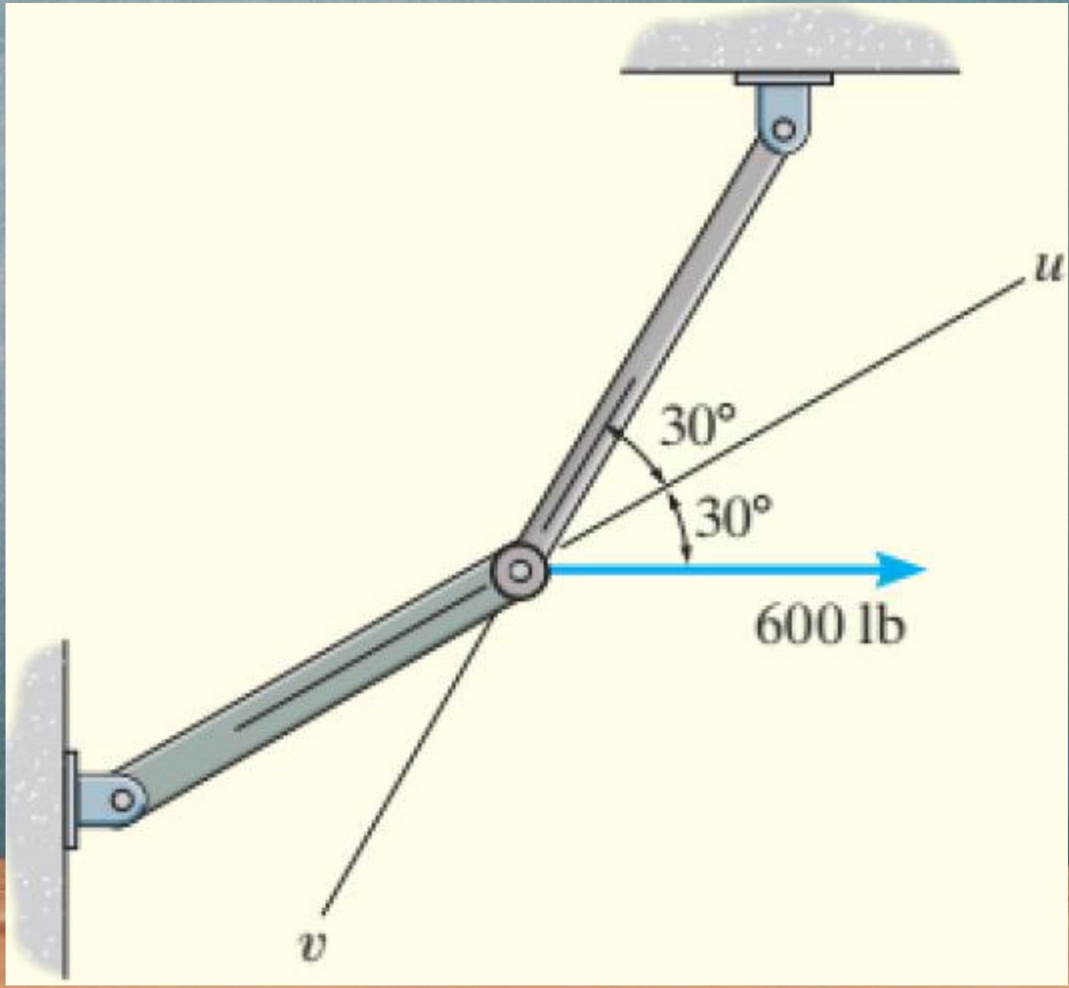
$T_{AC} = 144$ ~~lb~~ Kg

A NAG

Resolve the horizontal 600-lb force in Fig. into components acting along the u and v axes and determine the magnitudes of these components.

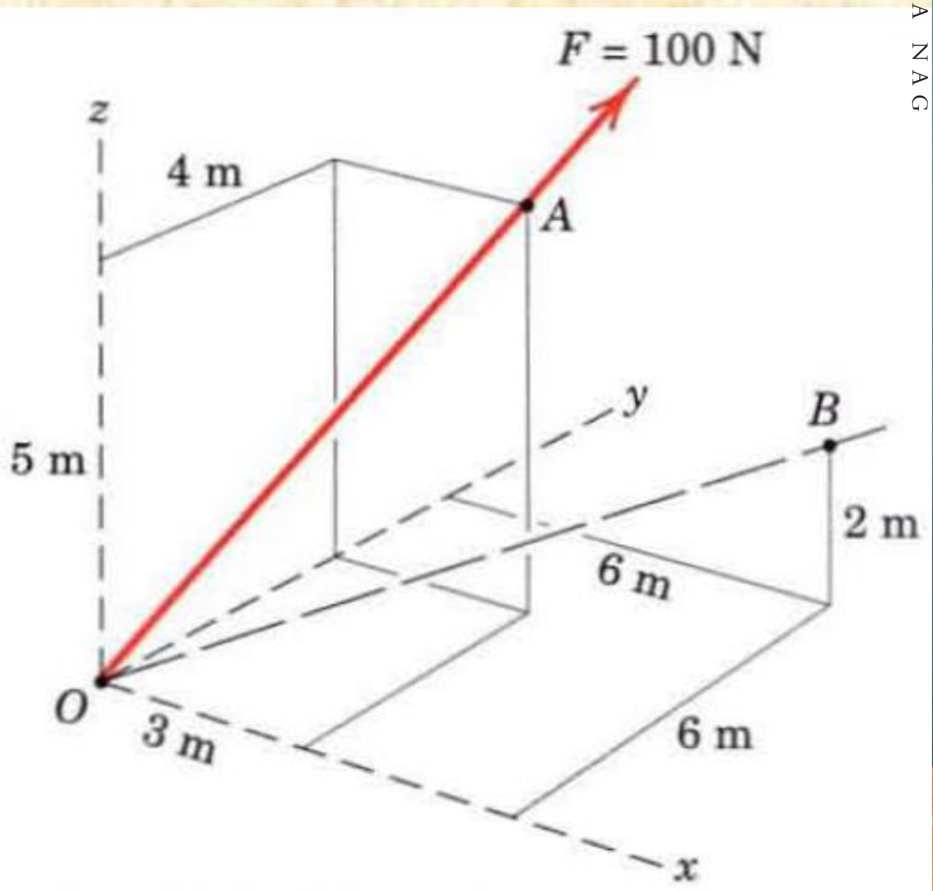
$F_u = 1039$ lb

$F_v = 600$ lb



A NAG

A force **F** with a magnitude of 100 N is applied at the origin O of the axes x-y-z as shown. The line of action of **F** passes through a point A whose coordinates are 3 m, 4 m, and 5 m. Determine (a) the x, y, and z scalar components of **F**, (b) the projection F_{xy} of **F** on the x-y plane, and (c) the projection F_{OB} of **F** along the line OB.



$F_x = 42.4\ \text{N} \qquad F_y = 56.6\ \text{N} \qquad F_z = 70.7\ \text{N}$

$F_{xy} = F\cos\theta_{xy} = 100(0.707) = 70.7\ \text{N}$

$\mathbf{n}_{OB} = 0.688\mathbf{i} + 0.688\mathbf{j} + 0.229\mathbf{k}$

$58.1\mathbf{i} + 58.1\mathbf{j} + 19.35\mathbf{k}$ N

A N A G

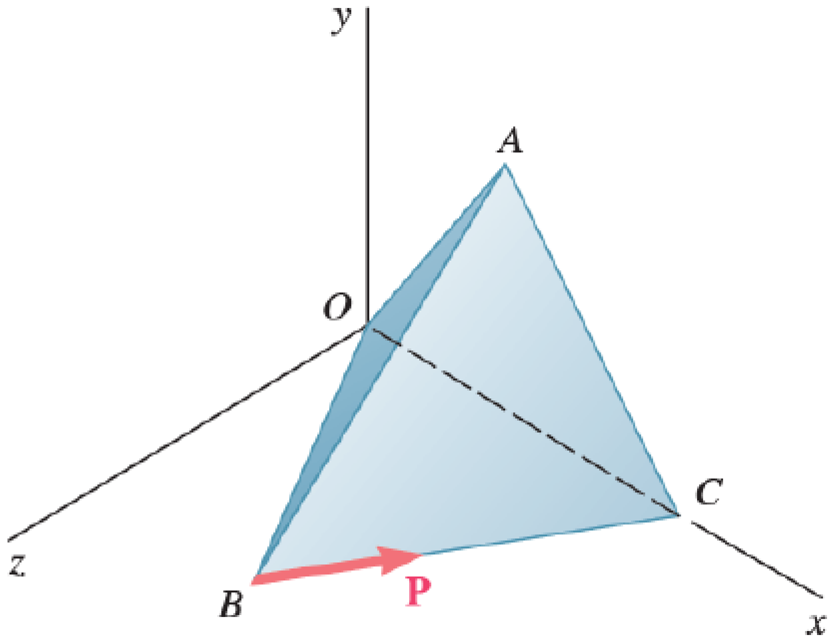
1. Determine the projection of the force P = 10i - 8j + 14k N on the directed line L which originates at point (2, -5, 3) and passes through point (5, 2, -4).

   Ans. -12.0 N

2. Find the cross product of P = 2.85i + 4.67j - 8.09k and Q = 28.3i + 44.6j + 53.3k. Ans. 610i-381j-5k

A NAG

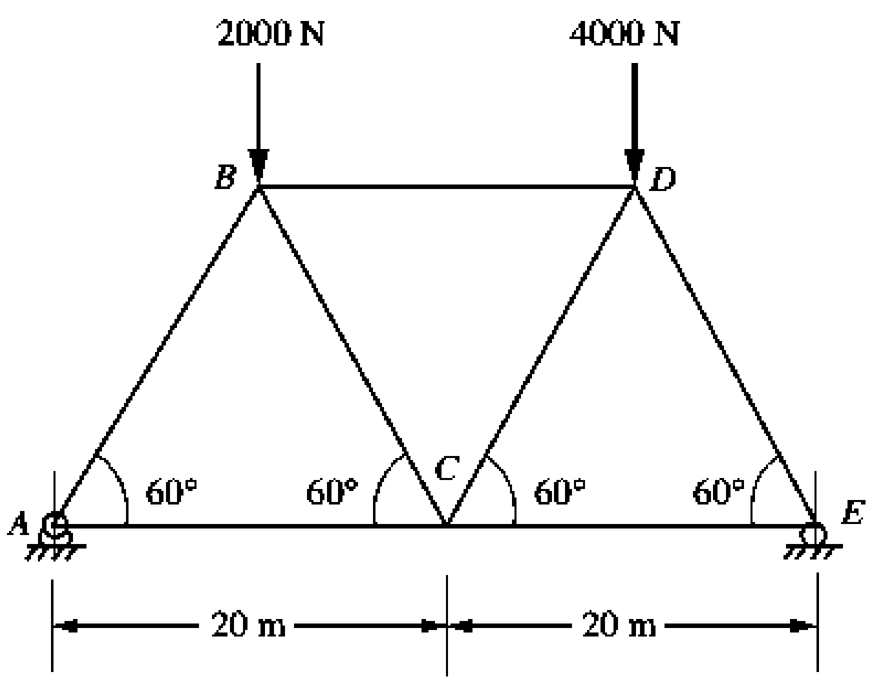
A regular tetrahedron has six edges of length a. A force $\mathbf{P}$ is directed as shown along edge BC. Determine the moment of $\mathbf{P}$ about edge OA.



$aP/\sqrt{2}$

A NAG

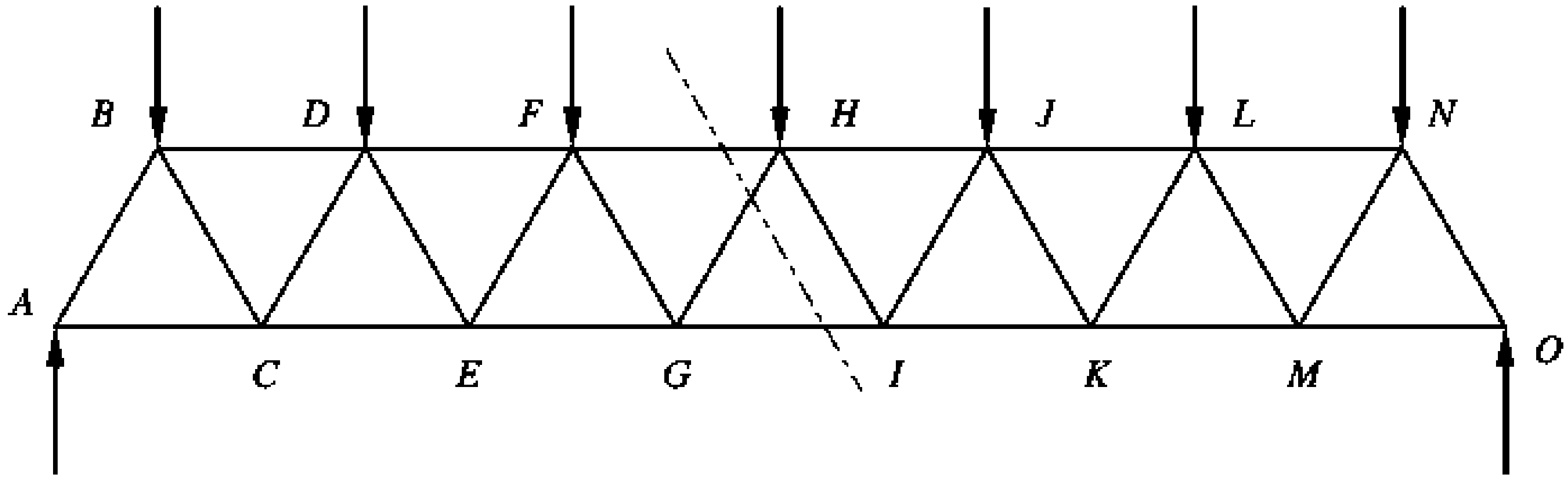
1. The simple triangle truss in Fig. supports two loads as shown. Determine reactions and the forces in each member.



$R_E = 3500$ N $R_A = 2500$ N $F_{BC} = 577$ N T. $F_{BD} = -1730$ N $F_{AB} = 2890$ N C, $F_{AC} = 1450$ N T.

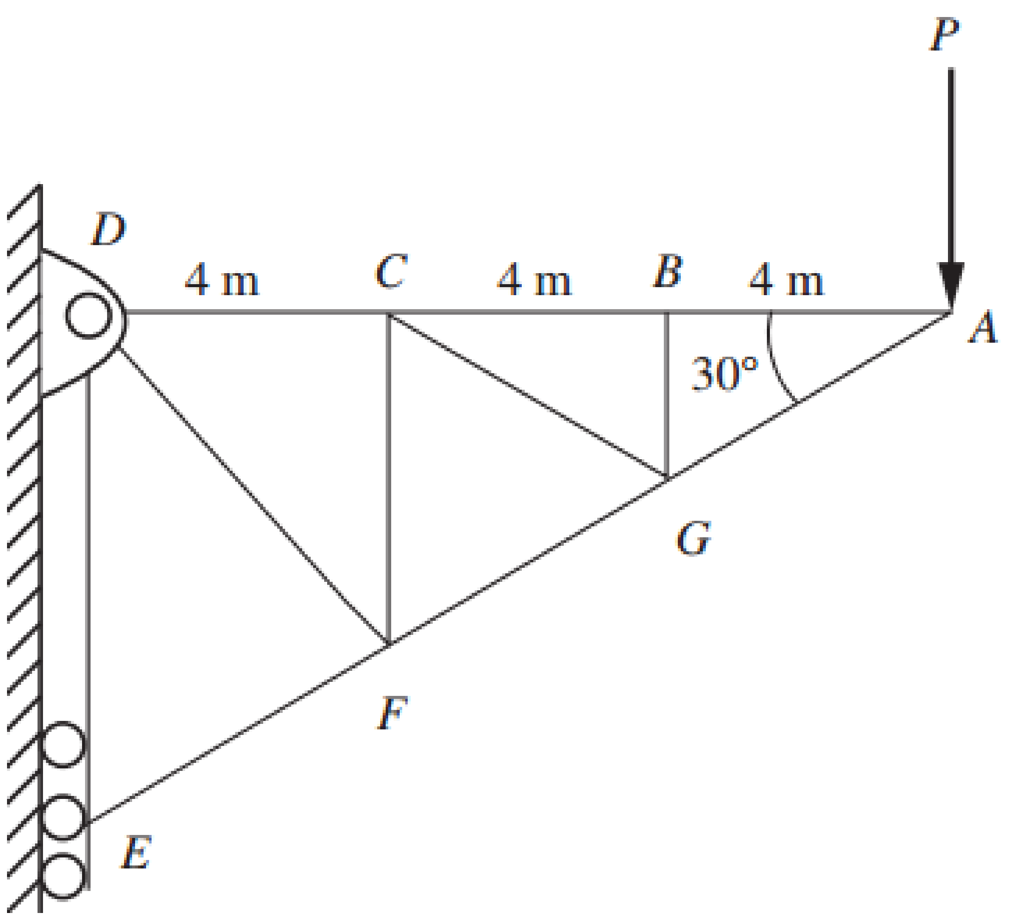
$F_{CD} = 577$ N C and $F_{CE} = 2020$ N T $F_{DE} = 4930$ N C and $F_{CE} = 2020$ N T.

2. Determine the forces in *FH*, *HG*, *IG*, and *IK* in the truss shown in Fig. Each load is 2 kN. All triangles are equilateral with sides of 4 m.



$F_{FH} = -13.9$ kN $\qquad F_{GI} = 14.4$ kN $\qquad F_{HG} = -1.15$ kN $\qquad F_{IK} = 13.3$ kN T

3. The maximum allowable force (tensile or compressive) in members *DC*, *DF*, or *EF* in the pin-connected truss shown in Fig. is known to be 160 kN. Determine the maximum permissible load *P*.

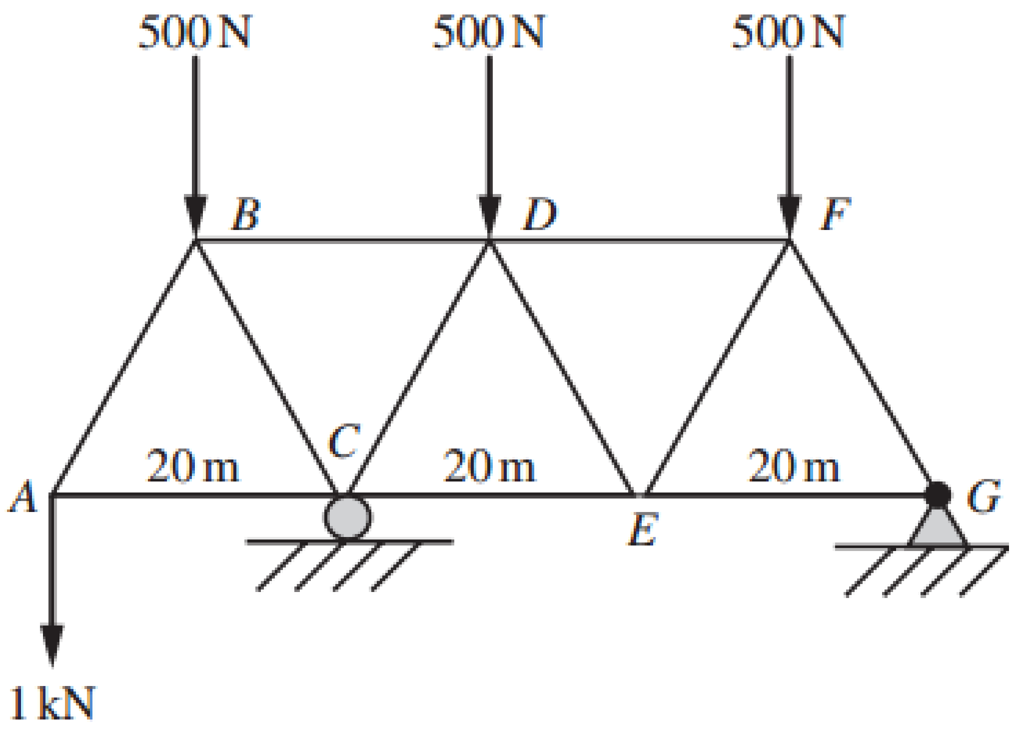


$$F_{EF} = -2P$$

$$F_{DC} = 1.73P$$

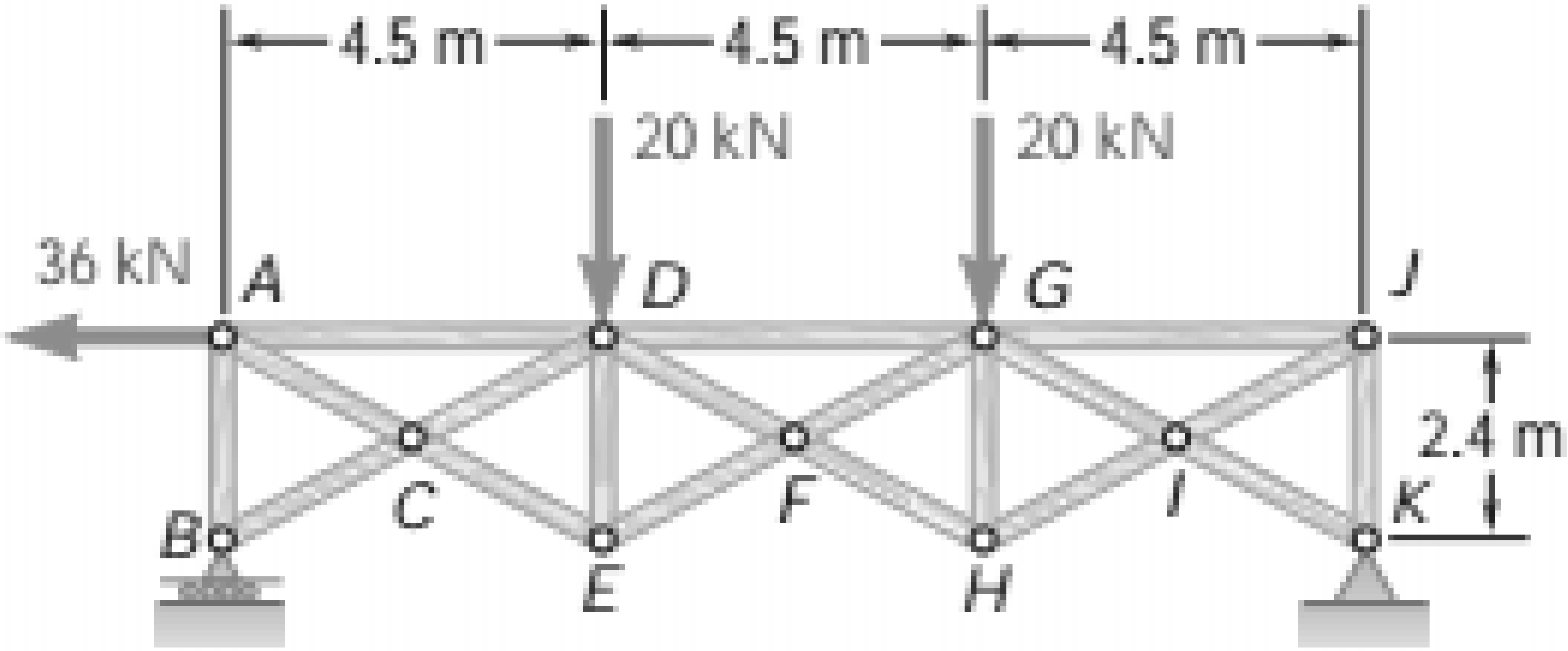
$$P = 80\ \text{kN}$$

4. Determine the forces in the members BD and DE of the truss in Fig. All triangles are equilateral.



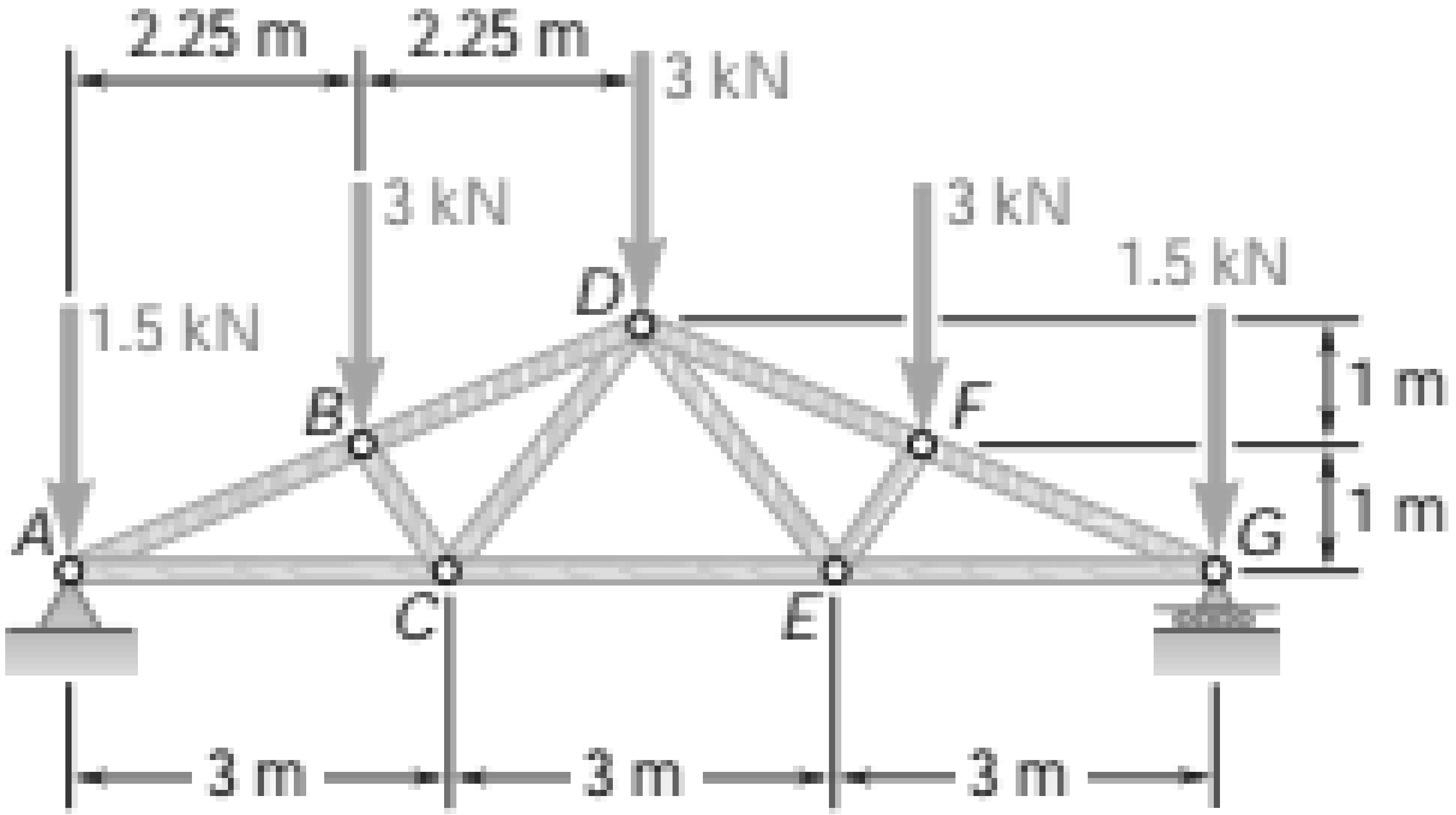
$R_C = 2625$ N $\quad F_{BD} = 1440$ N $\quad F_{CD} = -1300$ N

Determine the force in members DG, FG, and FH of the truss shown.



$\mathbf{B} = 26.4\text{ kN}\uparrow$; $\mathbf{K}_x = 36.0\text{ kN}\rightarrow$; $\mathbf{K}_y = 13.60\text{ kN}\uparrow$ $F_{DG} = 75.0\text{ kN}$ C $F_{FG} = 56.1\text{ kN}$ T $F_{FH} = 69.7\text{ kN}$ T

Determine the force in each member of the Fink roof truss shown. State whether each member is in tension or compression.



$A_y = G = 6.00\text{ kN}\uparrow$ $F_{AB} = 11.08\text{ kN}$ C $F_{AC} = 10.13\text{ kN}$ T $F_{BC} = 2.81\text{ kN}$ C $F_{BD} = 9.23\text{ kN}$ C

$F_{CD} = 2.81\text{ kN}$ T $F_{CE} = 6.75\text{ kN}$ T

$F_{DE} = F_{CD}$

$F_{DF} = F_{BD}$

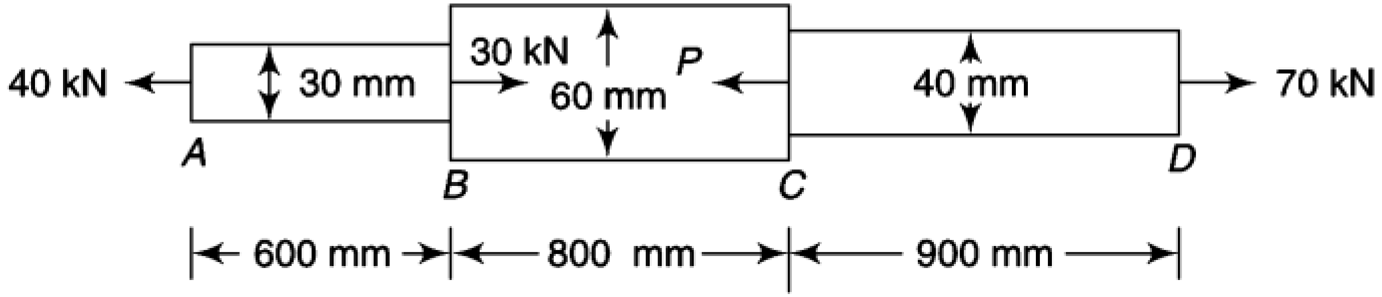
$F_{EF} = F_{BC}$

$F_{EG} = F_{AC}$

$F_{FG} = F_{AB}$

1.

A circular steel bar having three segments is subjected to various forces at different cross-sections as shown in Fig. 1.14. Determine the necessary force to be applied at Section *C* for the equilibrium of the bar. Also, find the total elongation of the bar. Take *E* = 202 GPa.



60 kN

0.4303 mm

2. A tension bar tapers from $(d + a)$ diameter to $(d - a)$ diameter. Prove that the error involved in using the mean diameter to calculate the Young's modulus is $(10a/d)^2$ percent.

Conical Section

Consider a bar of conical section under the action of an axial force P as shown in Fig. 1.18.

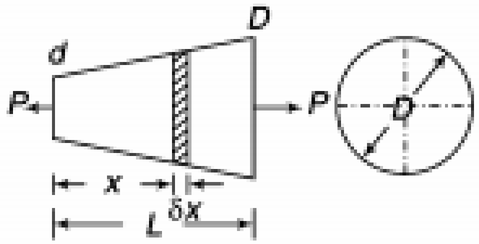


Fig. 1.18

Let

D = diameter at the larger end

d = diameter at the smaller end

L = length of the bar

E = Young's modulus of the bar material

Consider a very small length δx at a distance x from the small end.

The diameter at a distance x from the small end $= d + \dfrac{D-d}{L} \cdot x$

The extension of the small length $= \dfrac{P \cdot \delta x}{\dfrac{\pi}{4}\left(d + \dfrac{D-d}{L} x\right)^2 \cdot E}$ $\qquad \left(\Delta = \dfrac{PL}{AE}\right)$

Extension of the whole rod $= \displaystyle\int_0^L \frac{4P}{\pi(d + (D-d)x/L)^2 \cdot E} \cdot dx$

$$= \frac{4P}{\pi E}\int_0^L \left(d + \frac{D-d}{L} \cdot x\right)^{-2} \cdot dx = -\frac{4P}{\pi E} \cdot \frac{L}{(D-d)}\left(\frac{1}{(d + (D-d)x/L)}\right)_0^L$$

$$= \frac{4PL}{\pi E(D-d)}\left(\frac{1}{d} - \frac{1}{D}\right) = \frac{4PL}{\pi E(D-d)}\left(\frac{D-d}{dD}\right) = \frac{4PL}{\pi E d D} \qquad (1.7)$$

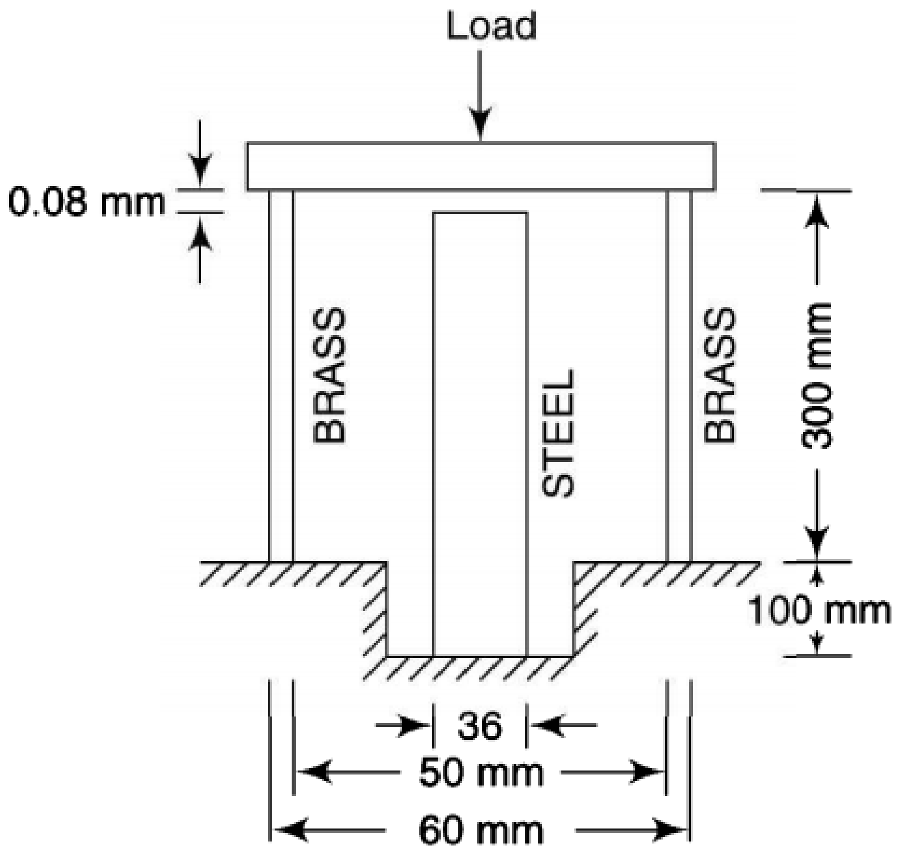
3.

A round steel rod supported in a recess is surrounded by a co-axial brass tube as shown in Fig. The level of the upper end of the rod is 0.08 mm below that of the tube. Determine

(i) the magnitude of the maximum permissible axial load which can be applied to a rigid plate resting on the top of the tube, the permissible values of the compressive stresses are 105 MPa for steel and 75 MPa for brass

(ii) the amount by which the tube is shortened by a load if the compressive stresses in the steel and the brass are the same

Take E_s = 210 GPa and E_b = 105 GPa.



136.555 kN

0.24 mm

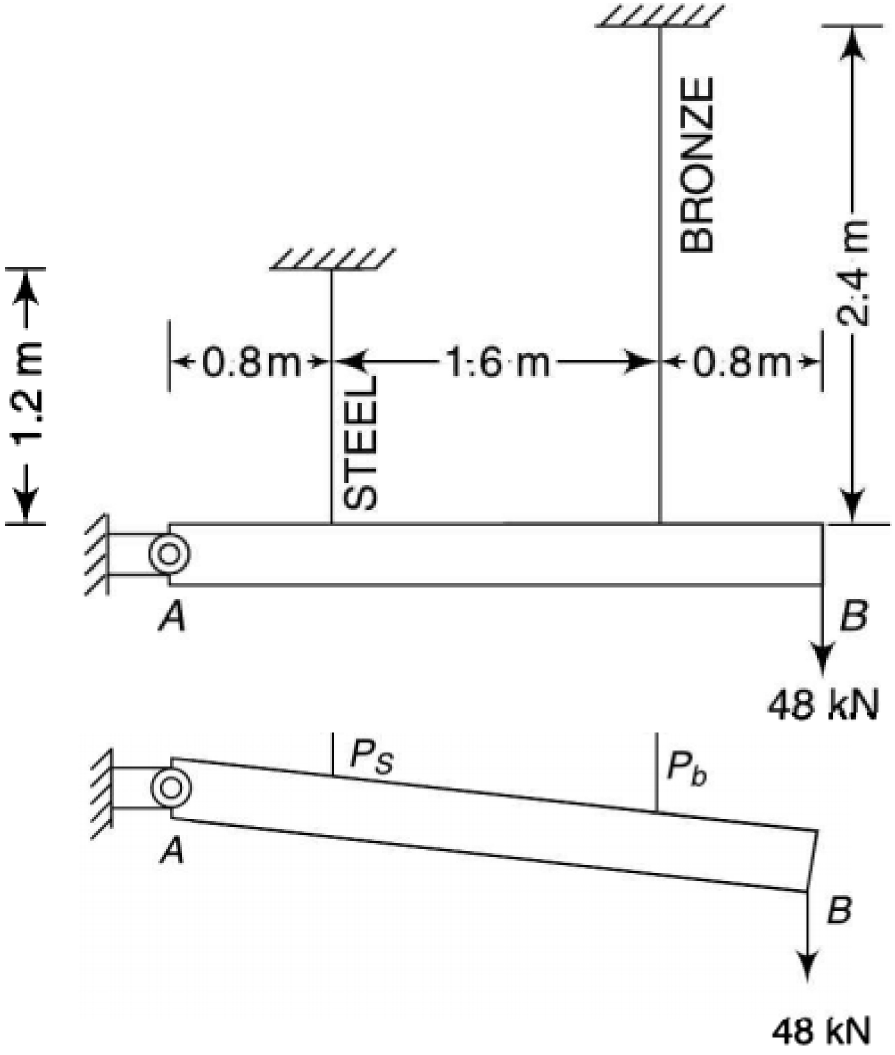
4.

A rigid horizontal bar *AB* hinged at *A* is supported by a 1.2-m long steel rod and a 2.4 m long bronze rod, both rigidly fixed at the upper ends (Fig.). A load of 48 kN is applied at a point that is 3.2 m from the hinge point *A*. The areas of cross-section of the steel and bronze rods are 850 mm² and 650 mm² respectively. Find

(i) the stress in each rod

(ii) the reaction at the pivot point

E_s = 205 GPa and E_b = 82 GPa



$$\sigma_b = \frac{37\,073}{650} = 57.04 \text{ MPa and}$$

$$\sigma_s = \frac{80\,781}{850} = 95.04 \text{ MPa}$$

69.854 kN

5.

A bar, 12 mm in diameter, is acted upon by an axial load of 20 kN. The change in diameter is measured as 0.003 mm. Determine

(i) the Poisson's ratio

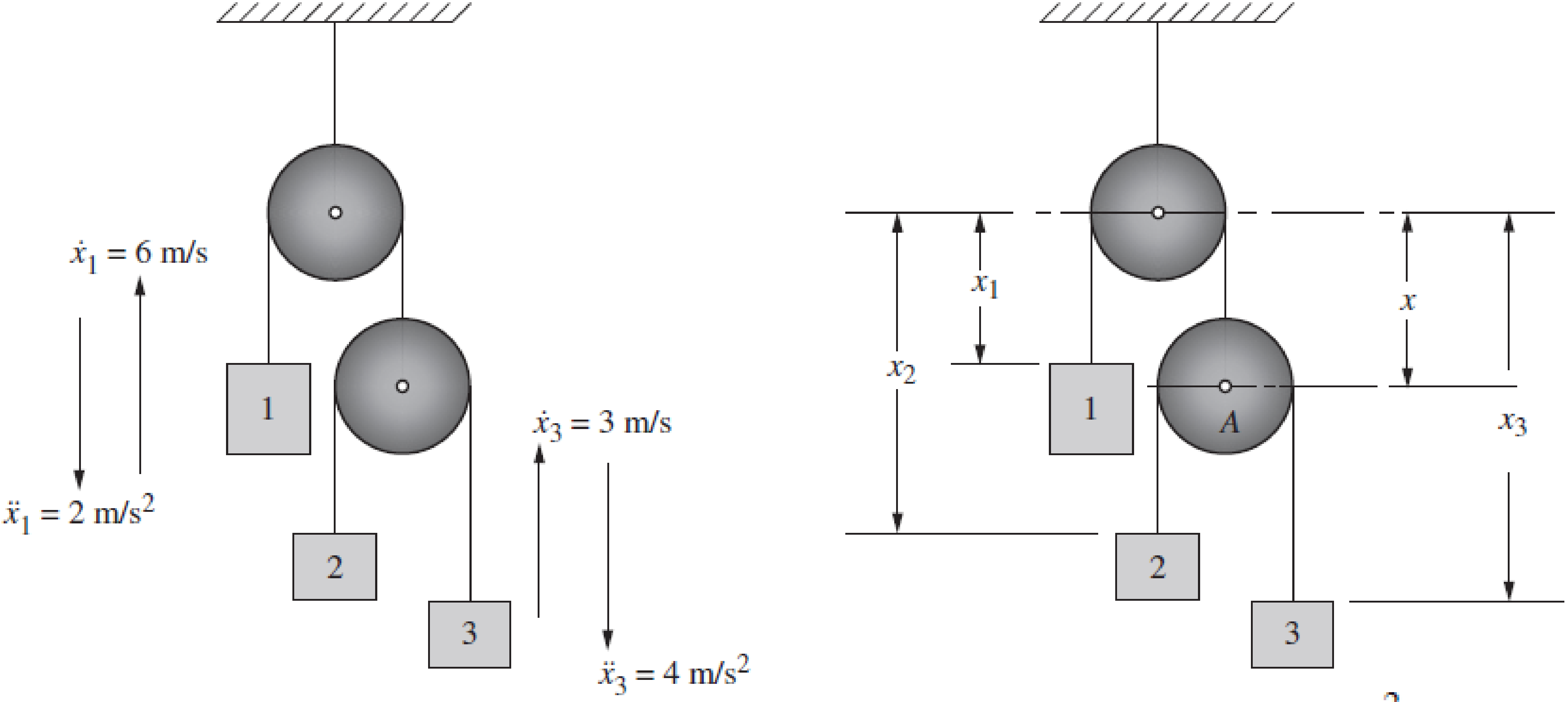
(ii) the modulus of elasticity and the bulk modulus

The value of the modulus of rigidity is 80 GPa.

$$\nu = 0.2923 \qquad K = 165\ 921 \text{ MPa}$$

1.

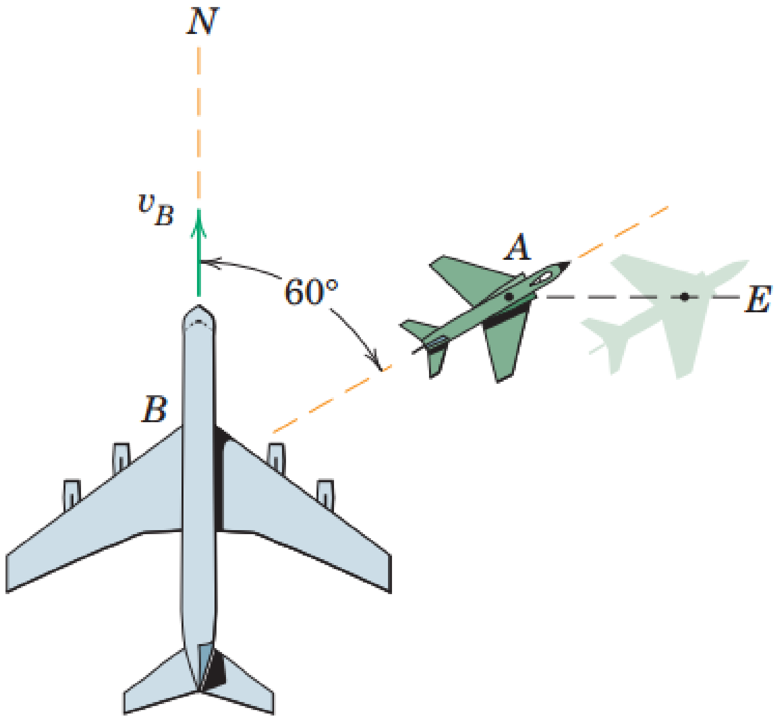
In the system sketched in Fig. 2-11(a), determine the velocity and acceleration of block 2 at the instant shown.



-15 m/s (down) 8 m/s^2

2.

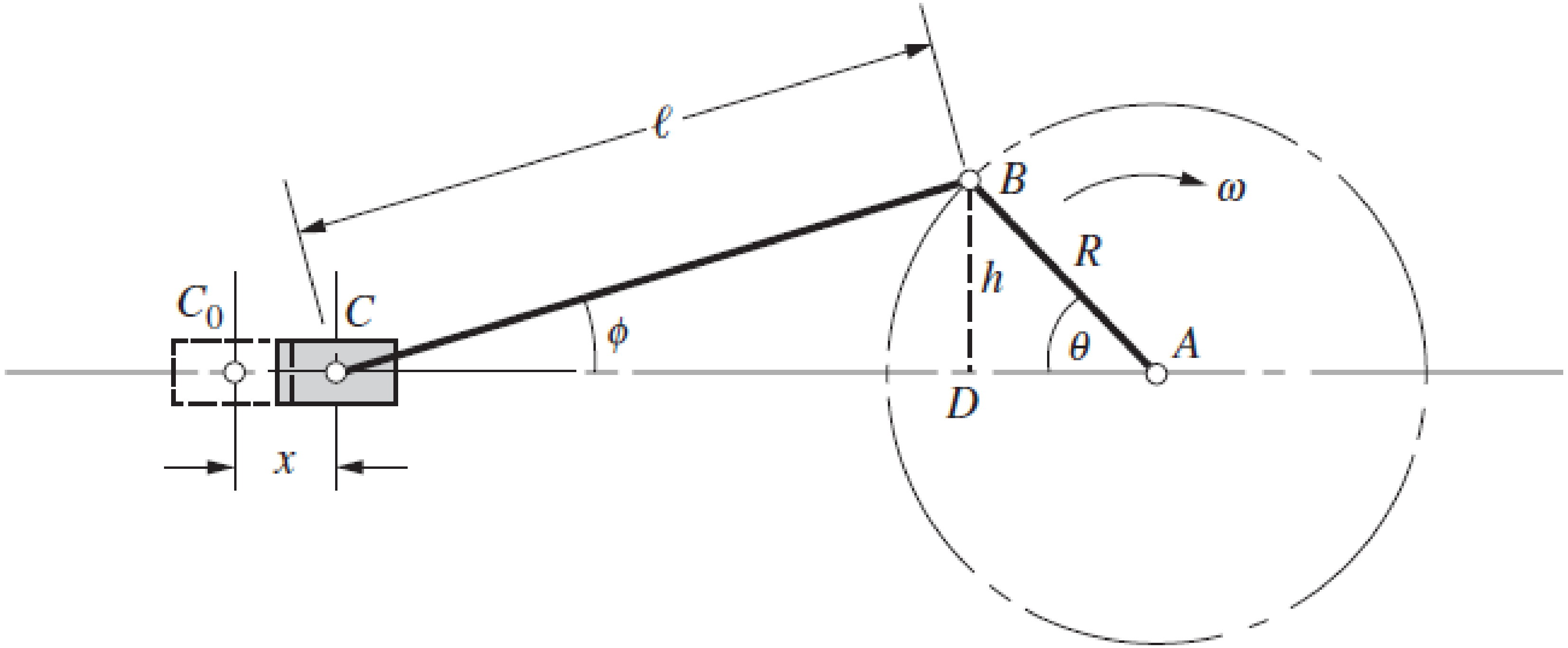
The jet transport B is flying north with a velocity $v_B = 600$ km/h when a smaller aircraft A passes underneath the transport headed in the 60° direction shown. To passengers in B, however, A appears to be flying sideways and moving east. Determine the actual velocity of A and the velocity which A appears to have relative to B.



$v_A = 1200$ km/h, $v_{A/B} = 1039$ km/h

3.

Determine the linear displacement, velocity, and acceleration of the crosshead C in the slider crank mechanism of Fig. 2-19 for any position of the crank R that is rotating at a constant angular velocity ω rad/s.



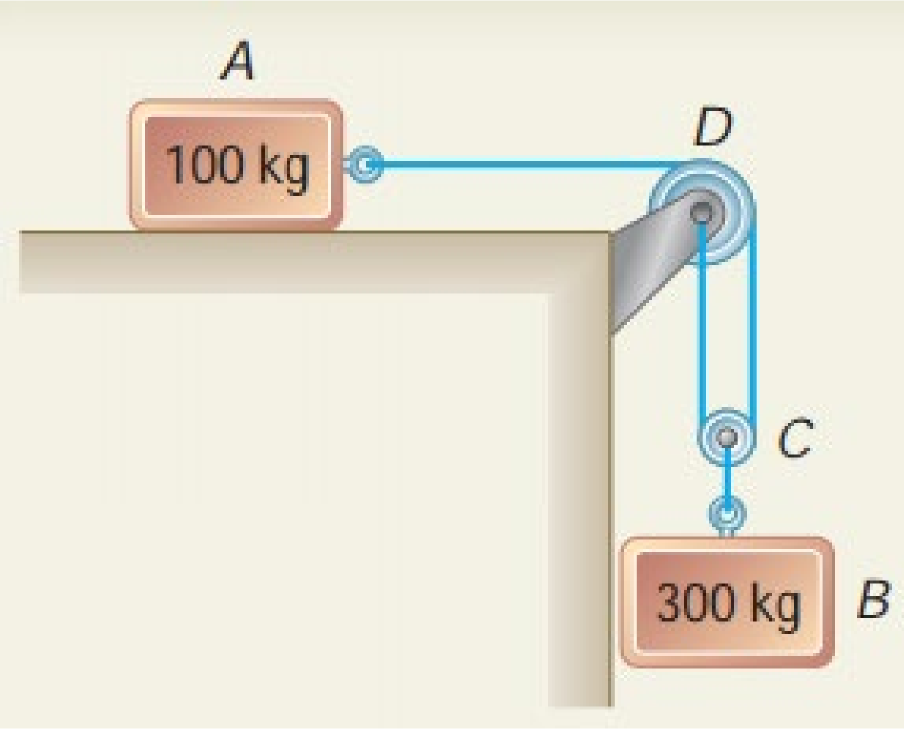
$$\boxed{R(1-\cos\theta)+\frac{R^2}{2\ell}\sin^2\theta}$$

$$\boxed{R\omega\left(\sin\theta+\frac{R}{2\ell}\sin 2\theta\right)}$$

$$\boxed{R\omega^2\left(\cos\theta+\frac{R}{\ell}\cos 2\theta\right)}$$

4.

The two blocks shown start from rest. The horizontal plane and the pulley are frictionless, and the pulley is assumed to be of negligible mass. Determine the acceleration of each block and the tension in each cord.



$a_A = 8.40\ \text{m/s}^2$

$a_B = 4.20\ \text{m/s}^2$

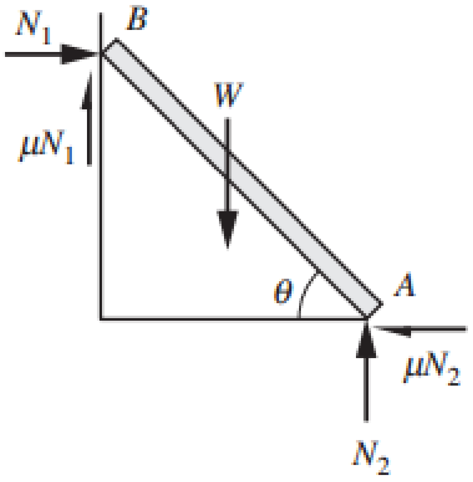
$T_1 = 840\ \text{N}$

$T_2 = 1680\ \text{N}$

An effort of 200 N is required just to move a certain body up an inclined plane of angle 15° the force acting parallel to the plane. If the angle of inclination of the plane is made 20° the effort required, again applied parallel to the plane, is found to be 230 N. Find the weight of the body and the coefficient of friction.

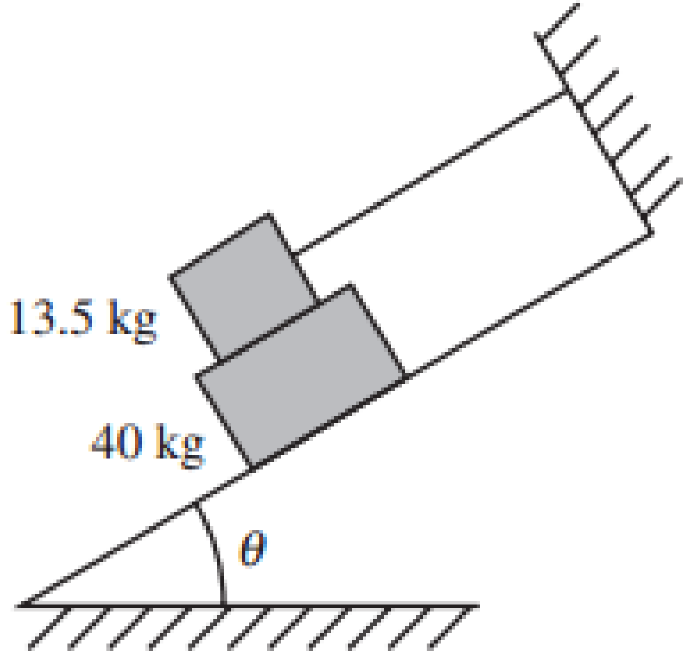
0.259 392.9 N

Determine the smallest angle θ for equilibrium of a homogeneous ladder of length L leaning against a wall. The coefficient of friction for all surfaces is μ.



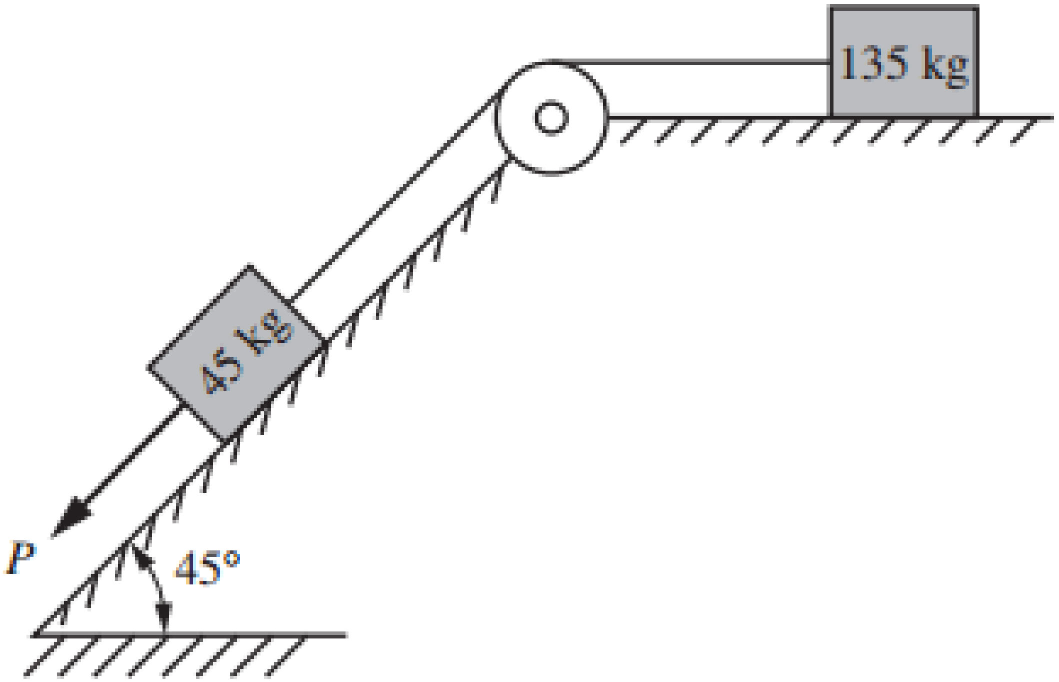
$$\theta = \tan^{-1}\frac{1-\mu^2}{2\mu}$$

What should be the value of the angle θ so that motion of the 40-kg block impends down the plane? The coefficient of friction μ for all surfaces is 0.333.



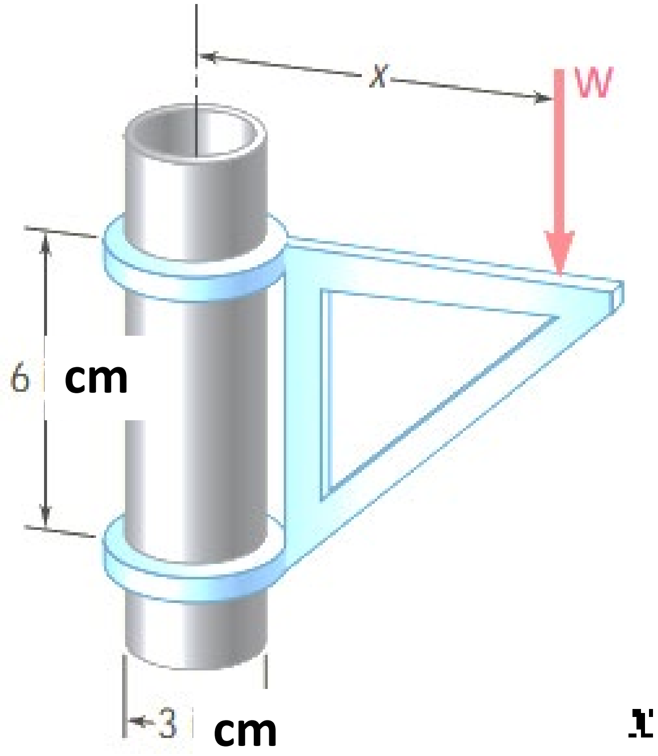
$\theta = 29.1°$.

Determine the necessary force P acting parallel to the plane to cause motion to impend. Assume that the coefficient of friction is 0.25 and that the pulley is smooth.



$P = 97.2\ \text{N}$

The movable bracket shown may be placed at any height on the 3 cm diameter pipe. If the coefficient of static friction between the pipe and bracket is 0.25, determine the minimum distance x at which the load **W** can be supported. Neglect the weight of the bracket.



$x = 12$ cm

Two blocks A and B, connected by a horizontal rod and frictionless hinges are supported on two rough planes as shown in Fig.

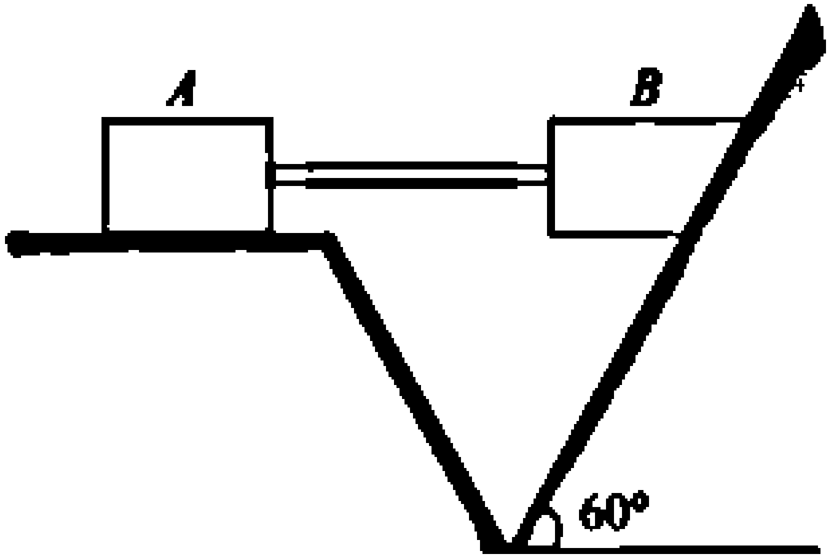


Fig.

The coefficients of friction are 0.3 between block A and the horizontal surface, and 0.4 between block B and the inclined surface. If the block B weighs 100 N, what is the smallest weight of block A, that will hold the system in equilibrium?

262.3 N

- ***Find the power transmitted by a belt running over a pulley of 600 mm diameter at 200 r.p.m. The coefficient of friction between the belt and pulley is 0·25, angle of lap 160° and maximum tension in the belt is 2·5 kN.***

$$P = (T_1 - T_2)\, v = (2{\cdot}5 - 1{\cdot}24)\, 2\pi = 7{\cdot}92 \quad \text{kW}$$

Screw Jack and Belt

1. The mean radius of the screw of a square threaded screw jack is 25 mm. The pitch of thread is 7.5 mm. If the coefficient of friction is 0.12, what effort applied at the end of lever 60 cm length is needed to raise a weight of 2 kN. ***Ans. 14.1N***

2. A laminated belt 8 mm thick and 150 mm wide drives a pulley of 1·2 m diameter at 180 r.p.m. The angle of lap is 190° and mass of the belt material is 1000 kg/m3. If the stress in the belt is not to exceed 1·5 N/mm2 and the coefficient of friction between the belt and the pulley is 0·3, determine the power transmitted when the centrifugal tension is (i) considered, and (ii) neglected. ***Ans. (i) 11·74 kW (ii) 12·84 kW***